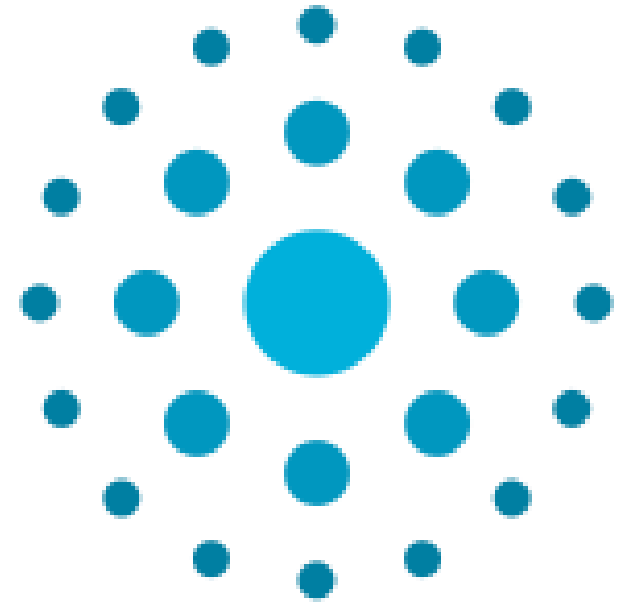


MEFISTO: Identifying Temporal and Spatial Patterns of Variation from Multimodal Data

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Introduction to MEFISTO

- MEFISTO (Multivariate Functional Inference with Structured Shrinkage and Temporal Smoothing) is a **Bayesian factor analysis model**.
- Designed to handle **longitudinal** and **spatially structured multimodal data**.
- Extends **MOFA (Multi-Omics Factor Analysis)** by incorporating **Gaussian Process priors**.
- Used in **genomics, healthcare, neuroscience, and environmental science**.

Introduction to MEFISTO

- • MEFISTO is an extension of MOFA that incorporates temporal structure in latent factors.
- • It models shared and group-specific temporal variation in multi-omics and microbiome data.
- • Designed for sparse and irregularly sampled longitudinal datasets.
- • Helps in clustering, imputation, trajectory inference, and enrichment analysis.

Introduction to MEFISTO

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Purpose of MEFISTO

- Goal:** Identify latent temporal and spatial structures in high-dimensional multimodal data.
- Challenges Addressed:**
 - Modeling **smooth variations** over time or space.
 - Capturing **structured correlations** in multimodal datasets.
 - Handling **missing values and noisy data** efficiently.

Key Concepts and Components

- • Factor analysis for dimensionality reduction
- • Integration of spatial and temporal dependencies
- • Applications in single-cell genomics and biomedical research

Advancements Over State-of-the-Art Methods

- **Traditional Factor Analysis Methods:**

- Assume independent samples (e.g., PCA, MOFA)
- Gaussian Process Factor Analysis (GPFA) can model time but is not optimized for multi-view data.
- Lack of spatio-temporal modeling

- **MEFISTO Improvements:**

- Incorporates continuous covariates (time, space)
- MEFISTO integrates both time-series and multi-view learning, improving accuracy and interpretability.
- Uses Gaussian processes for better data-driven alignment

MEFISTO Methodology

- 1. Latent Factor Model: Represents data using a set of hidden factors.
- 2. Temporal Gaussian Processes: Captures smooth temporal variations.
- 3. Multi-View Learning: Models dependencies across different data types.
- 4. Variational Inference: Optimizes model parameters efficiently.

Methodology - MEFISTO Framework

- **Observed Data Representation:**

- Y = Observed multimodal data
- Z = Latent factors
- W = Factor weights
- ϵ = Noise

heterogeneity. As input MEFISTO expects a collection of matrices, where each matrix $Y^{m,g}$ corresponds to a group $g=1,\dots,G$ and view $m=1,\dots,M$ with N_g samples in rows and D_m features in columns. Each sample is further characterized by a covariate $C^g \in \mathbb{R}^{C \times N_g}$ that represents, for example, temporal or spatial coordinates. The matrices are jointly decomposed as

$$Y^{m,g} = Z^g W^{mT} + \epsilon^{m,g} \quad m = 1, \dots, M, g = 1, \dots, G,$$

where $Z^g \in \mathbb{R}^{N_g \times K}$ contains the K latent factors and $W^m \in \mathbb{R}^{D_m \times K}$ contains their weights. A feature- and view-wise sparsity prior is used for W^m as in previous multimodal factor analysis models^{7,8}. Unlike existing factor models, however, the

- **Gaussian Process Priors:** Ensures smooth transitions in latent factors.
- **Structured Shrinkage Priors:** Prunes irrelevant latent factors.
- **Bayesian Inference:** Estimates the model parameters.

Temporal & Spatial Modeling in MEFISTO

- Uses **Gaussian Process (GP) priors** to enforce smoothness in latent trajectories.

model additionally accounts for the covariate $\mathbf{C}^g$. Each factor value z_{nk}^g is modeled as a realization of a Gaussian process

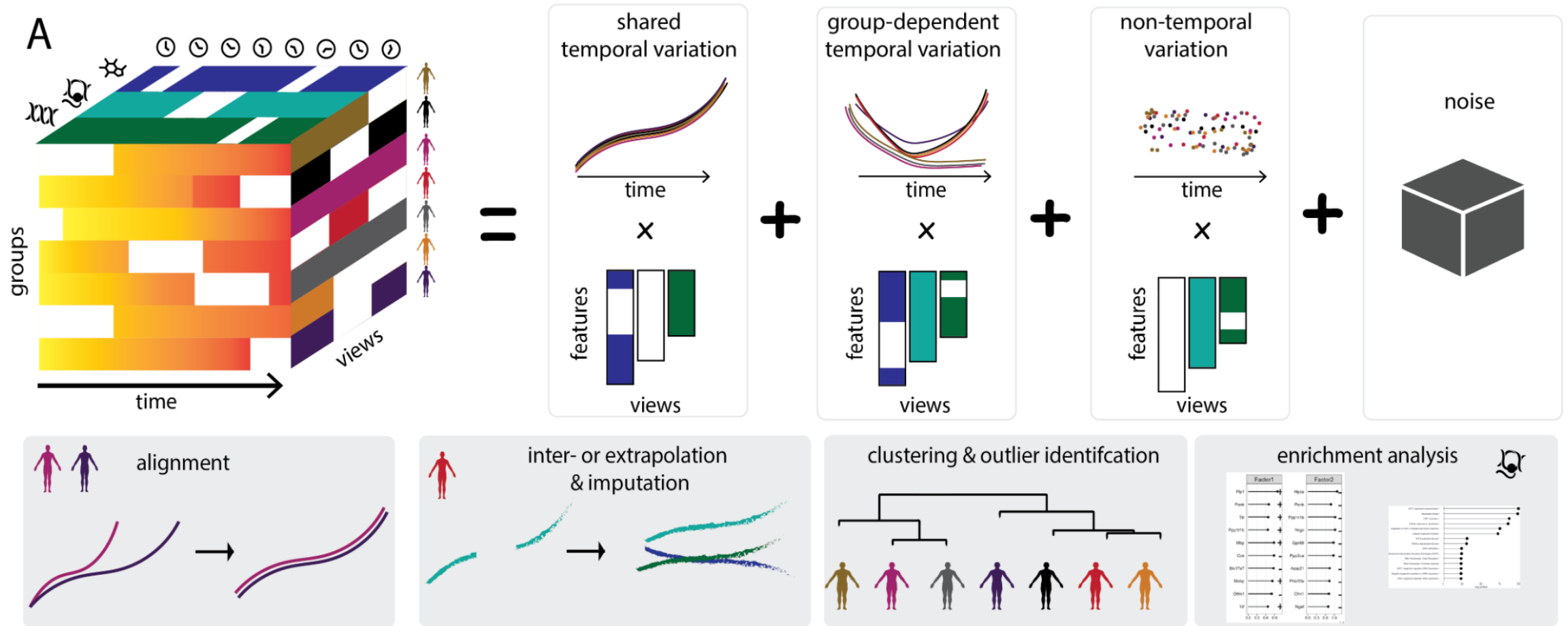
$$z_{nk}^g = f_k(\mathbf{c}_n^g) + \eta_{nk}^g \quad \text{with } f_k \sim \text{GP}(0, \kappa_k),$$

- GP covariance function:

$$k(t, t') = \sigma^2 \exp\left(-\frac{(t - t')^2}{2l^2}\right)$$

- l (length scale) controls smoothness over time.
- Learns **shared and modality-specific factors** across multiple data sources.
- Uses Gaussian processes to model spatial-temporal dependencies
 - Incorporates prior knowledge to enhance interpretability

MEFISTO Model Representation



Implementation and Tools

- • Implemented in Python using the MOFA+ framework
- • Supports integration with TensorFlow and PyTorch
- • Open-source and available for research applications

Advantages of MEFISTO

- • Handles missing data with interpolation.
- • Captures both shared and group-specific temporal dynamics.
- • Works well with multi-omics, microbiome, and clinical datasets.
- • Enhances clustering and outlier detection.

Results from the Paper:

Identifying temporal variation in EvoDevo data with MEFISTO

```
datadir = "C:/Users/praball/Documents/GitHub/UCA-M2-SEMESTER2/omics/MEFISTO/13233860"
✓ 0.0s

data = pd.read_csv(f"{datadir}/evodevo.csv", sep=",", index_col=0)
data
✓ 5.6s
```

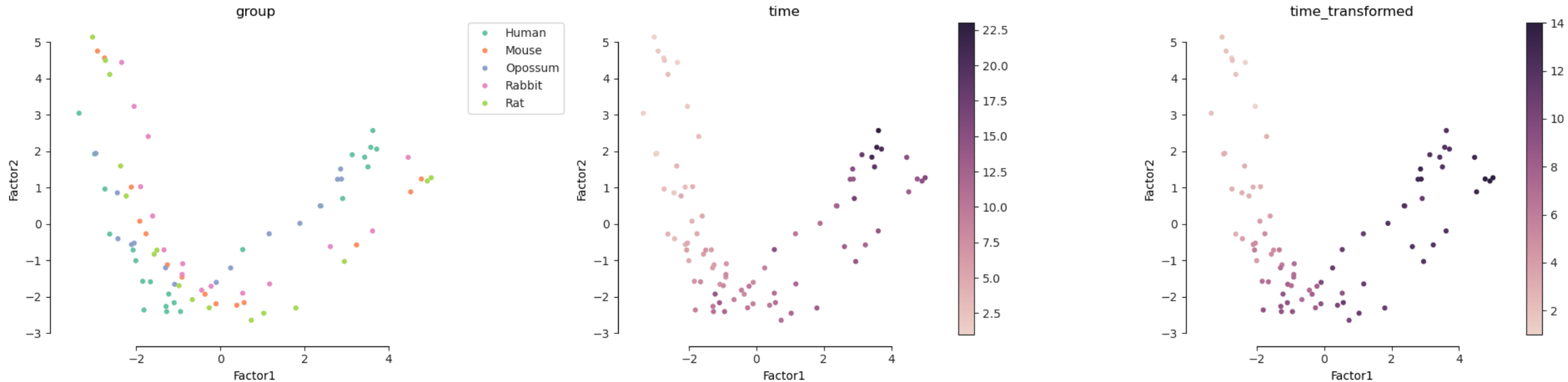
	group	view	sample	feature	value	time
1	Human	Brain	10wpc_Human	ENSG00000000457_Brain	8.573918	7
2	Human	Brain	10wpc_Human	ENSG00000001084_Brain	8.875957	7
3	Human	Brain	10wpc_Human	ENSG00000001167_Brain	11.265237	7
4	Human	Brain	10wpc_Human	ENSG00000001461_Brain	7.374965	7
5	Human	Brain	10wpc_Human	ENSG00000001561_Brain	7.311018	7
...	...	...	...	...	...	...
3193836	Human	Testis	youngMidAge_Human	ENSG00000271503_Testis	1.178014	21
3193837	Human	Testis	youngMidAge_Human	ENSG00000271601_Testis	1.178014	21
3193838	Human	Testis	youngMidAge_Human	ENSG00000272442_Testis	4.476201	21
3193839	Human	Testis	youngMidAge_Human	ENSG00000272886_Testis	1.178014	21
3193840	Human	Testis	youngMidAge_Human	ENSG00000273079_Testis	8.311393	21

3193840 rows × 6 columns

Results from the Paper:

Identifying temporal variation in EvoDevo data with MEFISTO

```
# Visualize factor space
mofax.plot_factors_scatter(m, x=0, y=1,
color=["group", "time", "time_transformed"])
```



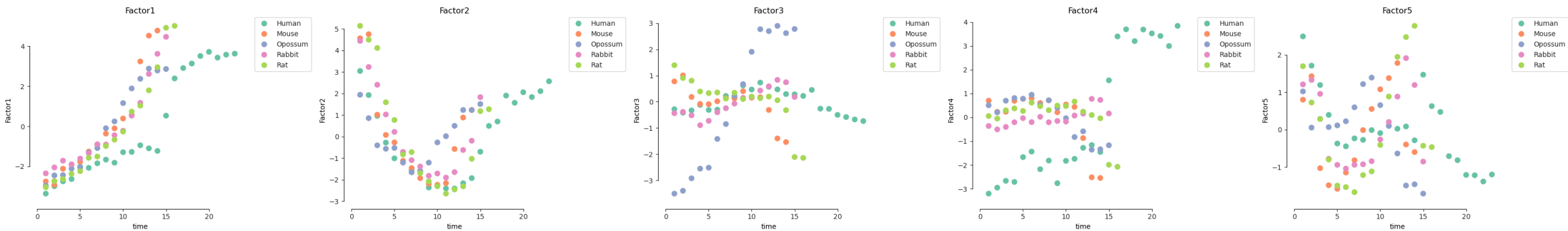
•Dataset Applications:

- Evolutionary gene expression atlas (5 species, 5 organs, temporal resolution)

Results from the Paper: Identifying temporal variation in EvoDevo data with MEFISTO

Latent factors versus common developmental time
We can plot the latent processes along the
inferred common developmental time.

Before alignment:



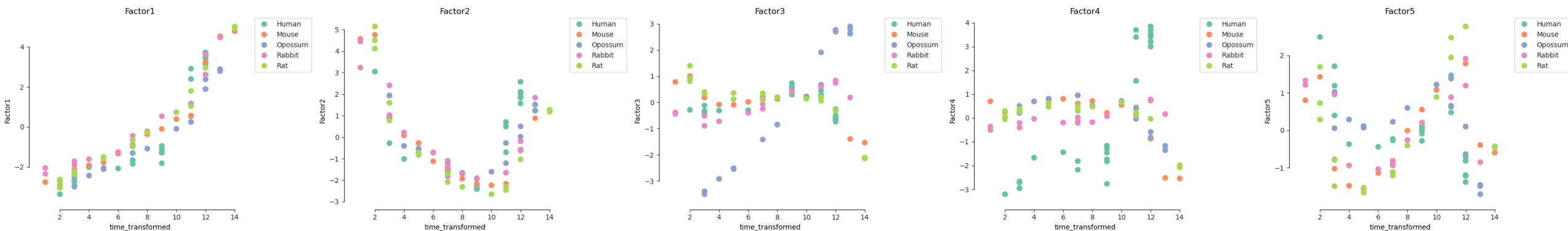
```
mofax.plot_factors_scatter(m, x="time",  
y=range(m.n_factors),  
ncols=5, size=70);
```


Results from the Paper: Identifying temporal variation in EvoDevo data with MEFISTO

After alignment:

-

```
# Plot latent factors over developmental time
mofax.plot_factors_scatter(m,
x="time_transformed", y=range(m.nfactors),
ncols=5, size=70)
```

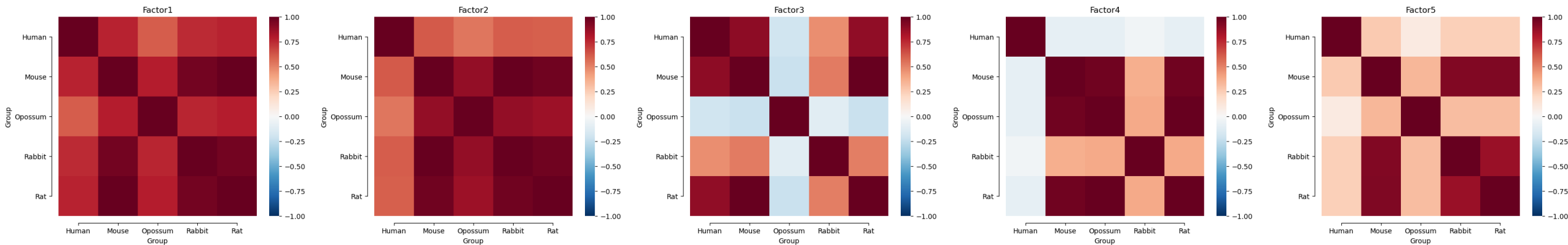


Results from the Paper: Identifying temporal variation in EvoDevo data with MEFISTO

Sharedness

The group kernel of the Gaussian process can give us insights into the extent to which a temporal pattern is shared across species for the developmental module captured by each process.

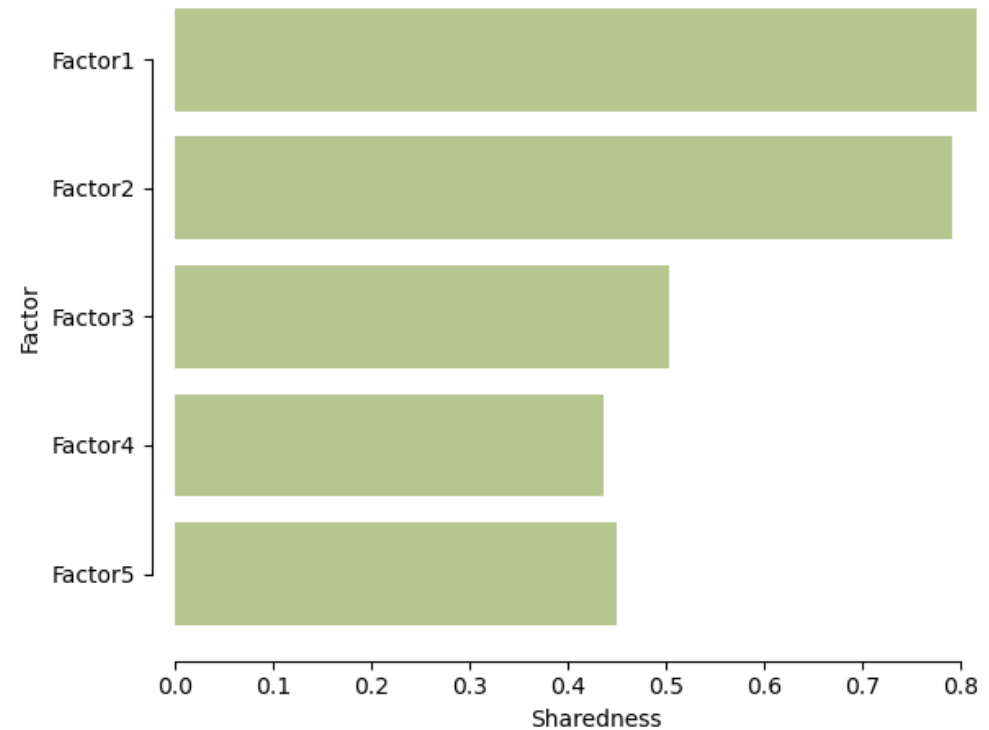
- Factors (factors 1, 2, and 5) seem to describe processes conserved across species while the 3RD and 5th factors show species-specific clusters:



Results from the Paper: Identifying temporal variation in EvoDevo data with MEFISTO

plot sharedness as a feature for each factor (fully
coloured bar means fully shared):

-
- `mofax.plot_sharedness(m);`



ON SOME RANDOM DATASET FROM T3

```
datadir = "C:/Users/praba/Documents/GitHub/UCA-M2-SEMESTER2/omics/MEFISTO/13233860"
data = pd.read_csv(f"{datadir}/gene_sample.csv", sep=",", index_col=0)
data
```

✓ 0.1s Python

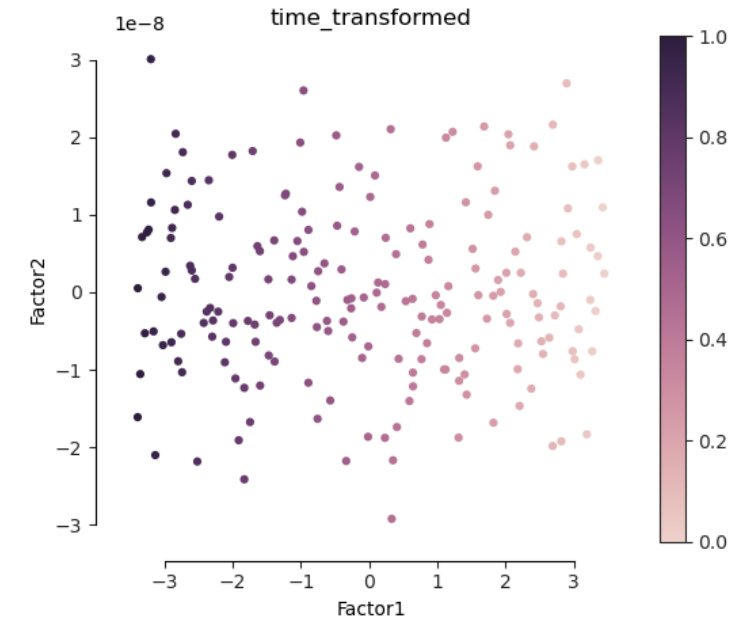
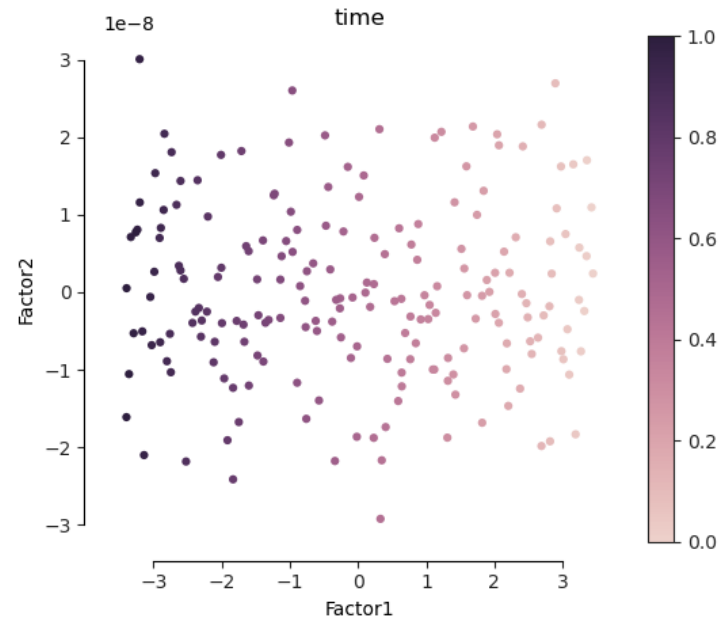
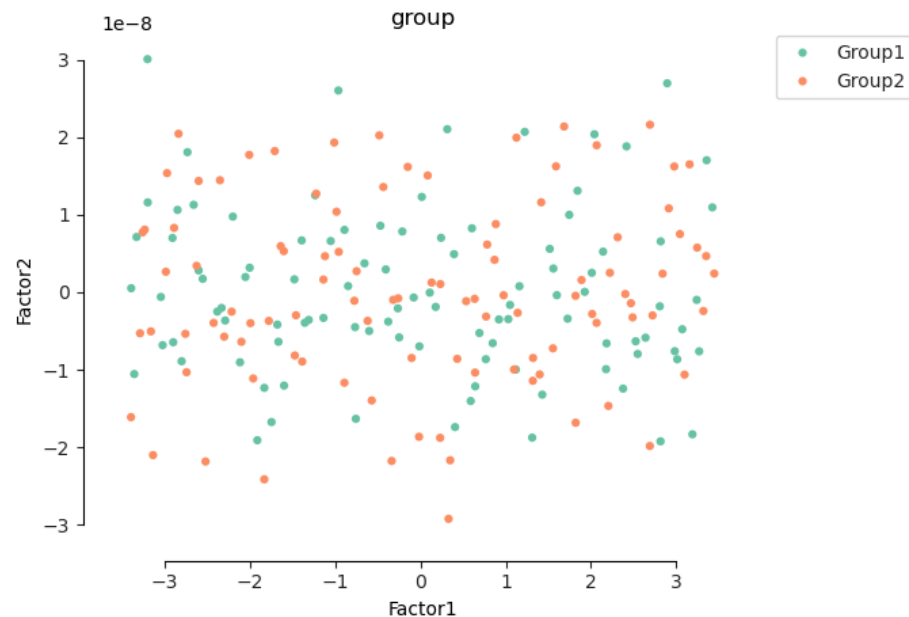
Gene_8	Gene_9	Gene_10	Gene_11	...	Gene_995	Gene_996	Gene_997	Gene_998	Gene_999	Gene_1000	group	time	view	sample
-1.278961	1.222139	-0.053799	-0.712353	...	-2.110290	-0.768964	-0.764824	-0.409113	-1.894955	0.663783	Group1	0.000000	Expression	Group1_Sample_1
-0.911009	1.776799	0.410499	-0.184165	...	-1.684041	-0.120163	0.217516	-0.828713	-2.004047	0.641395	Group1	0.010101	Expression	Group1_Sample_2
-0.724633	1.059882	-0.313065	-0.466676	...	-2.373904	-0.327907	-0.025051	-0.576952	-1.793250	-0.093917	Group1	0.020202	Expression	Group1_Sample_3
-1.053191	1.717510	-0.566020	-1.457603	...	-2.281203	-0.945741	-0.976907	-0.604897	-0.160841	-0.120119	Group1	0.030303	Expression	Group1_Sample_4
-0.889406	1.400257	-0.314301	-0.266859	...	-1.796331	0.539780	0.656982	-1.618957	-1.545181	-0.207028	Group1	0.040404	Expression	Group1_Sample_5
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
-1.408602	0.037874	-0.971069	0.210962	...	-0.937408	2.038015	0.391774	0.530516	-0.820411	-0.729341	Group2	0.959596	Expression	Group2_Sample_96
-0.095258	-0.472068	-0.195430	-0.556027	...	-0.132237	1.755551	1.444109	1.262918	0.406020	-0.793913	Group2	0.969697	Expression	Group2_Sample_97
-1.221107	-0.949336	-1.575395	0.617981	...	-1.350264	1.410991	1.202119	0.842615	0.207602	-0.210831	Group2	0.979798	Expression	Group2_Sample_98
-1.031002	-0.596403	-1.171168	0.639837	...	-0.769122	1.759926	1.752244	0.551708	0.809047	-0.015323	Group2	0.989899	Expression	Group2_Sample_99
-0.708601	-0.200129	-0.720689	-0.064303	...	-0.880636	1.537068	0.275423	-0.683958	0.199720	-1.283541	Group2	1.000000	Expression	Group2_Sample_100

data_long							
✓	0.0s						
	group	view	sample	time	feature	value	
0	Group1	Expression	Group1_Sample_1	0.000000	Gene_1	-0.002430	
1	Group1	Expression	Group1_Sample_2	0.010101	Gene_1	-0.914753	
2	Group1	Expression	Group1_Sample_3	0.020202	Gene_1	-0.914762	
3	Group1	Expression	Group1_Sample_4	0.030303	Gene_1	-1.437530	
4	Group1	Expression	Group1_Sample_5	0.040404	Gene_1	-1.043454	
...	...	...	...	...	...	...	...
199995	Group2	Expression	Group2_Sample_96	0.959596	Gene_1000	-0.729341	
199996	Group2	Expression	Group2_Sample_97	0.969697	Gene_1000	-0.793913	
199997	Group2	Expression	Group2_Sample_98	0.979798	Gene_1000	-0.210831	
199998	Group2	Expression	Group2_Sample_99	0.989899	Gene_1000	-0.015323	
199999	Group2	Expression	Group2_Sample_100	1.000000	Gene_1000	-1.283541	
200000 rows × 6 columns							

ON SOME RANDOM DATASET FROM T3

```
# Visualize factor space
mofax.plot_factors_scatter(m, x=0, y=1,
color=["group", "time", "time_transformed"])
```

BEFORE ALIGNMENT:

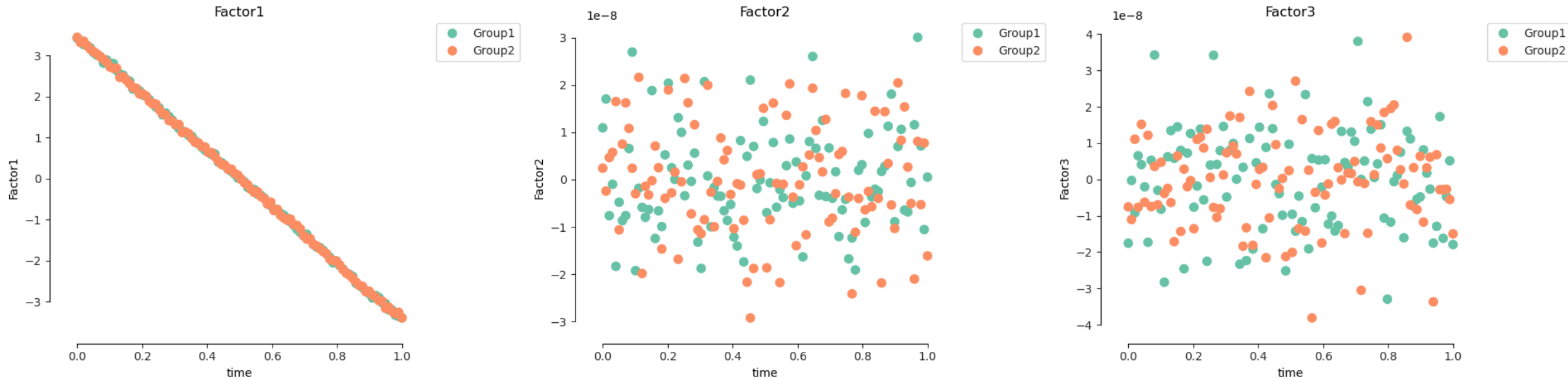


ON SOME RANDOM DATASET FROM T3

```
# Plot a scatter plot of the latent factors  
versus the time covariate.
```

```
mofax.plot_factors_scatter(m, x="time",  
y=range(m.nfactors), ncol=3, size=70)  
plt.show()
```

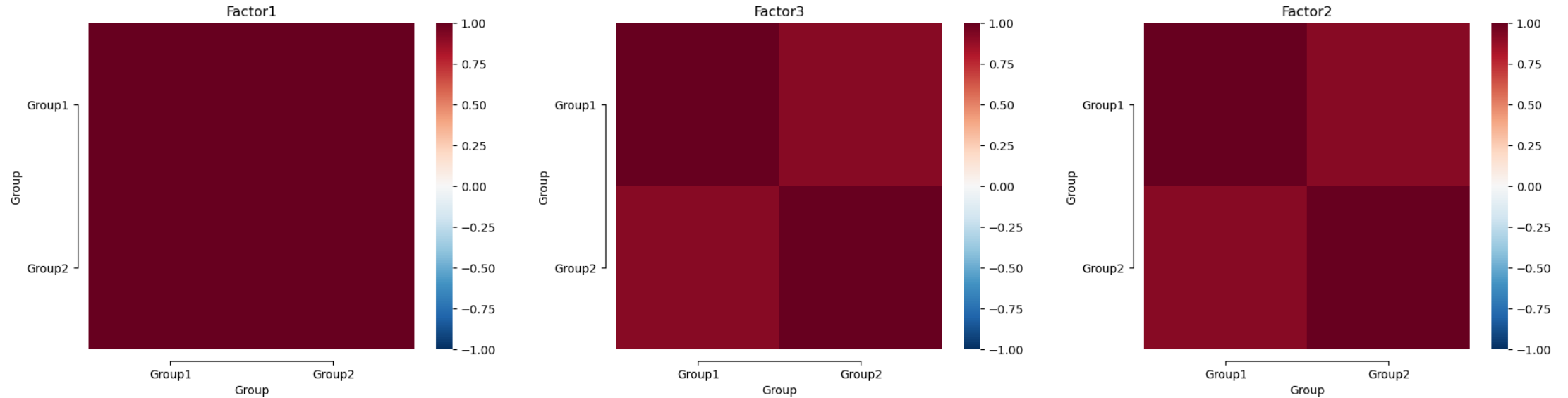
AFTER ALIGNMENT



- The variance explained plot (R^2) shows how much of the gene expression variability is captured by each factor.
- The scatter plot of latent factors vs. time should show a smooth trend:

ON SOME RANDOM DATASET FROM T3

```
# Plot sharedness across species  
to_plot = ["Factor1", "Factor2", "Factor3"]  
mofax.plot_group_kernel(m, factors=to_plot)  
  
plt.show()
```



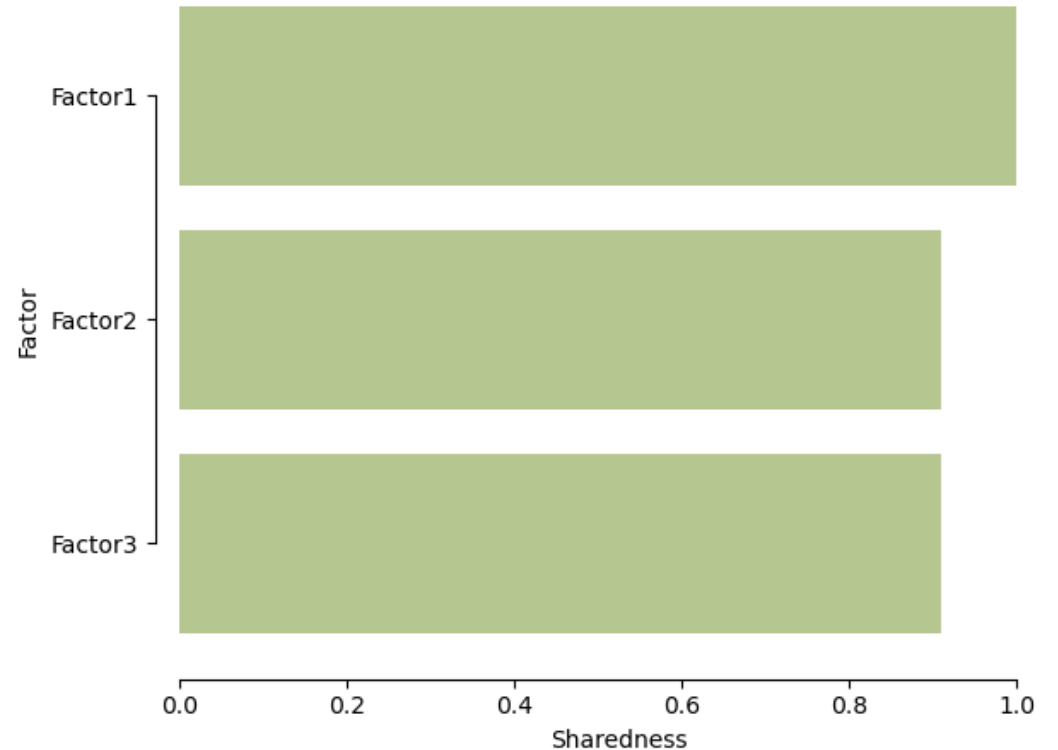
ON SOME RANDOM DATASET FROM T3

plot sharedness as a feature for each factor (fully coloured bar means fully shared):

-

```
mofax.plot_sharedness(m);
```

-



Conclusion

- • MEFISTO is a powerful method for analyzing sparse, multi-view longitudinal data.
- • Combines temporal Gaussian processes with factor models.
- • Enables improved clustering, imputation, and enrichment analysis.
- • A valuable tool for multi-omics and microbiome research.

Thank You